

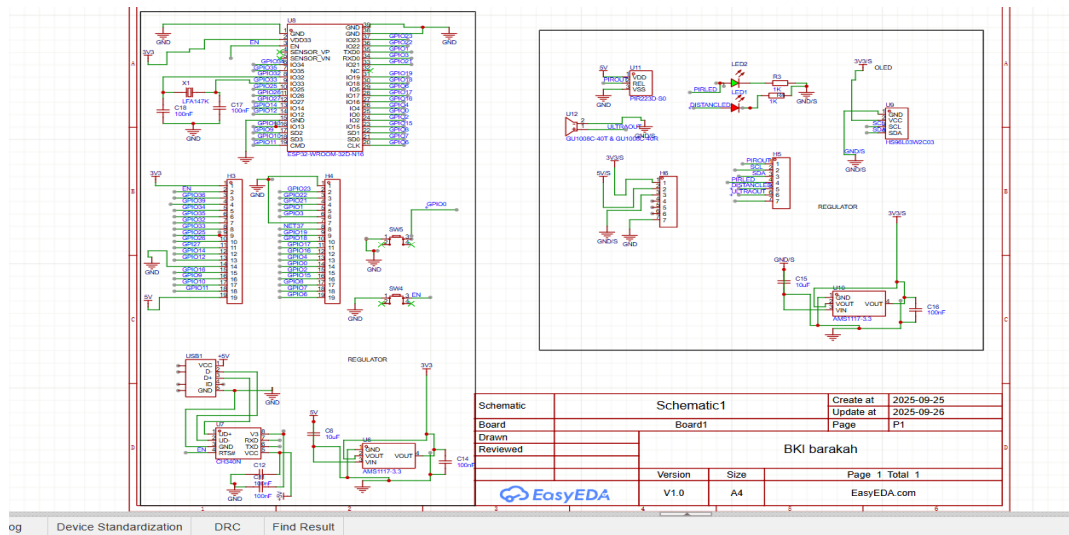
Project Documentation

Project Name: Smart IoT Sensor System

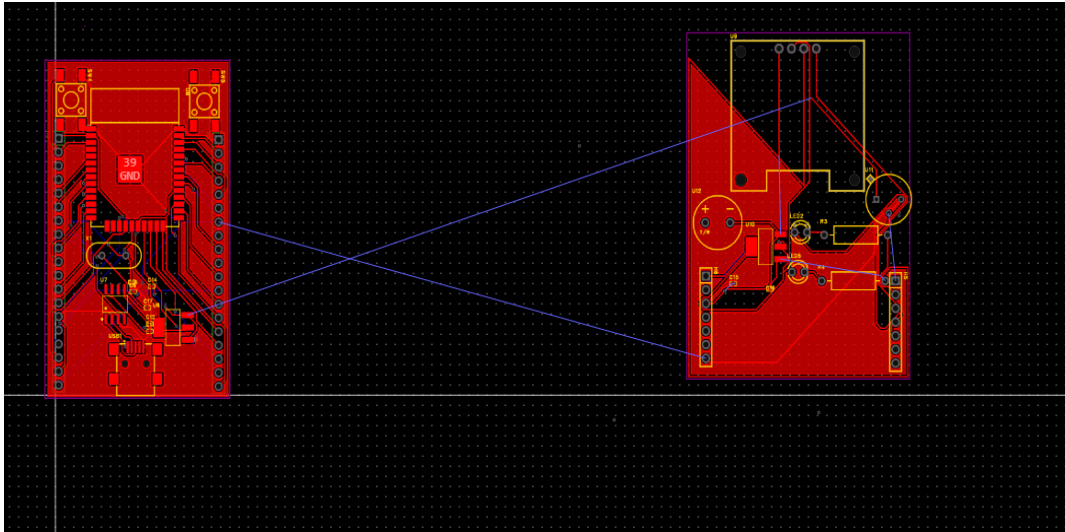
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This document provides the full documentation of the designed electronic project. The project is developed using EasyEDA and contains schematic diagrams, PCB layouts, 3D views, and a detailed Bill of Materials (BOM).

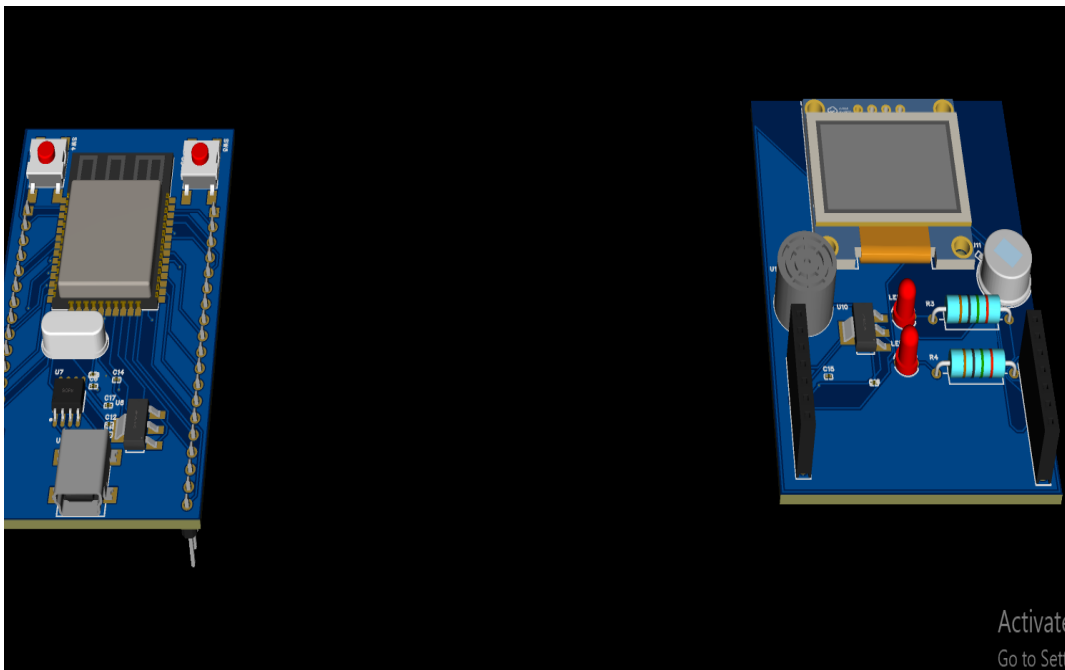
3D PCB Views



PCB Layout



Circuit Schematic



Bill of Materials (BOM)

Designator	Part	Value/Type	Footprint	Quantity
U8	ESP32 Module	ESP32-WROOM-32D-N16	Module	1
U4	Voltage Regulator	AMS1117-3.3	SOT-223	1
U7	USB-Serial IC	CH340/CH340N	SSOP/SOIC	1
USB1	USB Type-C Connector	Power/Programming	USB-C	1

X1	Crystal/Oscillator	LFA147K	SMT	1
U11	PIR Sensor	PIR223D-SD	Module	1
U12	Ultrasonic Sensor	HC-SR04 type	Transducer	1
U9	OLED Display	0.96" SSD1306	I2C Module	1
Buzzer	Piezo Buzzer	Passive/Active	TH	1
LED1,LED2	LED	3mm Red	LED-3mm	2
R3,R4	Resistor	1k Ω	0805/0603	2
C15,C16...	Capacitors	10 μ F, 100nF	0805/0603	several
H3,H4,H5,H6	Pin Headers	2.54mm	Header	multiple

The listed components form the complete project, combining sensing (PIR, ultrasonic), display (OLED), control (ESP32), power regulation (AMS1117), user interaction (buttons, LEDs), and connectivity (USB-C, headers).

This project demonstrates a complete embedded system design workflow starting from schematic design, PCB layout, and 3D visualization to BOM documentation. It is applicable for IoT, security, and automation systems.